Lilly Sweet

(913) 335-2048 • lillysweet@berkeley.edu • LinkedIn • Portfolio • Secret Security Clearance

EDUCATION

University of California, Berkeley MS/PhD in Mechanical Engineering Aug 2025 – Dec 2026

GPA: 3.92 **Graduate Coursework:** Continuum, Rheology, Adv. CFD & Heat Transfer, Adv. Thermo, Adv. Fluids

University of Kansas BS in Mechanical Engineering Jan 2021 – May 2025

GPA: 3.98 *(with distinction)* **Involvement:** Aero Co-Team Lead: Formula SAE

WORK EXPERIENCE

SpaceX Aero/Thermal Engineer Hawthorne, CA June 2026 – Aug 2026

- Conducting aerodynamics/thermal analysis for the Falcon Rocket, Dragon Capsule, Starfall, and special projects.

SpaceX Falcon Graduate Engineer Hawthorne, CA June 2026 – Aug 2026

- Executed FUN3D CFD simulations to obtain pressure and heat maps for a quick disconnect aero ramp and V4 Mini tile cover. Ran thermal analysis using CFD heat maps to confirm/size TPS systems for both hardware.
- Preprocessed, executed, and post-processed FUN3D CFD simulations simulating the wind tunnel experiments across multiple angles of attack. Found high corroboration between wind tunnel and CFD results. Tared out the CFD's sting impacts on aerodynamic coefficients, compared against clean Starfall CFD and flight data, finding high agreement.
- Developed a Newtonian theory MATLAB script to analyze stability on simplified Starshield satellite models, enabling aero-stability analysis across multiple configurations to assess atmospheric demise.

SpaceX Raptor Graduate Engineer Hawthorne, CA May 2025 – Aug 2025

- Integrated an **AI camera system** to automate machining process inspections, eliminating human-eye subjectivity and achieving **5–20× faster** throughput with improved repeatability and reliability. Utilized RobotStudio.
- **Automated** ingestion and organization of **nozzle extension test data** into an MQTT-backed database, enabling engineers to rapidly query results and validate propulsion hardware quality.

Tesla Controls Engineering Intern Fremont, CA May 2024 – Aug 2024

- Led a **PLC migration** effort covering >$200K in hardware and labor, minimizing downtime while modernizing controls infrastructure for long-term reliability.

Burns & McDonnell Mechanical Engineering Intern Kansas City, MO May 2023 – Aug 2023

- Developed **automated engineering workflow tools** by combining Python and PowerShell scripting, enabling unattended execution of complex analysis routines previously performed manually or in Excel.

US Army Combat Medic Sergeant (M-Day) Jan 2020 – Jan 2026

- **Security Clearance:** Secret **Rank:** Sergeant (E-5) (Non-Commissioned Officer) **Honorable Discharge**

ENGINEERING PROJECTS

- **n^{th} Order FVM Euler Solver:** Used spectral, FR/CPR, and DG methods to achieve n^{th} order accuracy on 1D Euler flow through a converging/diverging nozzle.
- **2D Viscous with Unstructured Meshing FV Solver:** A custom solver that simulates supersonic flow over a NACA0012. Roe's flux, Superbee & MC limiters (2^{nd} order w/ LS) were able to successfully resolve shocks.
- **FSAE Aerodynamics:** Ran iterative STAR-CCM+ simulations. Improved the team's process for higher fidelity simulations. Mechanically designed and manufactured mounting and wings.

SKILLS

- **Analysis:** STAR-CCM+, Patran/Nastran, Ansys (Mechanical & Fluent), Thermal Desktop, Pointwise, HPC
- **Coding:** Python, MATLAB, C++/Arduino, Machine Learning
- **CAD/General:** SolidWorks, NX, ABB Robot Studio, PLC's (ABB, Siemens), Ignition, Carbon Fiber Layups